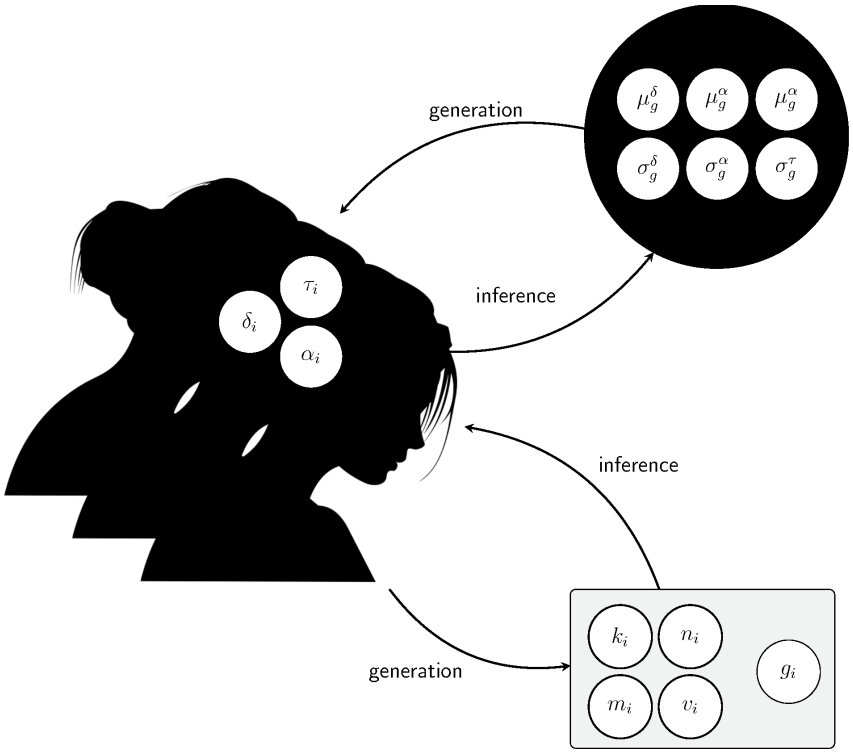
The Hierarchical EZ Diffusion Model

Michael D. Lee

Hierarchical EZ Diffusion Model Structure

- The hierarchical EZ diffusion model has the structure
  - observed number of decisions n_i, correct response time mean m_i, and correct response time variance v_i for each individual
  - drift rate δ_i, boundary separation α_i, and non-decision time τ_i parameters for each individual
  - group indicators g_i and group level Gaussians with means μ_g^δ, μ_g^α, and μ_g^τ, and standard deviations σ_g^δ, σ_g^α, and σ_g^τ



Numerosity

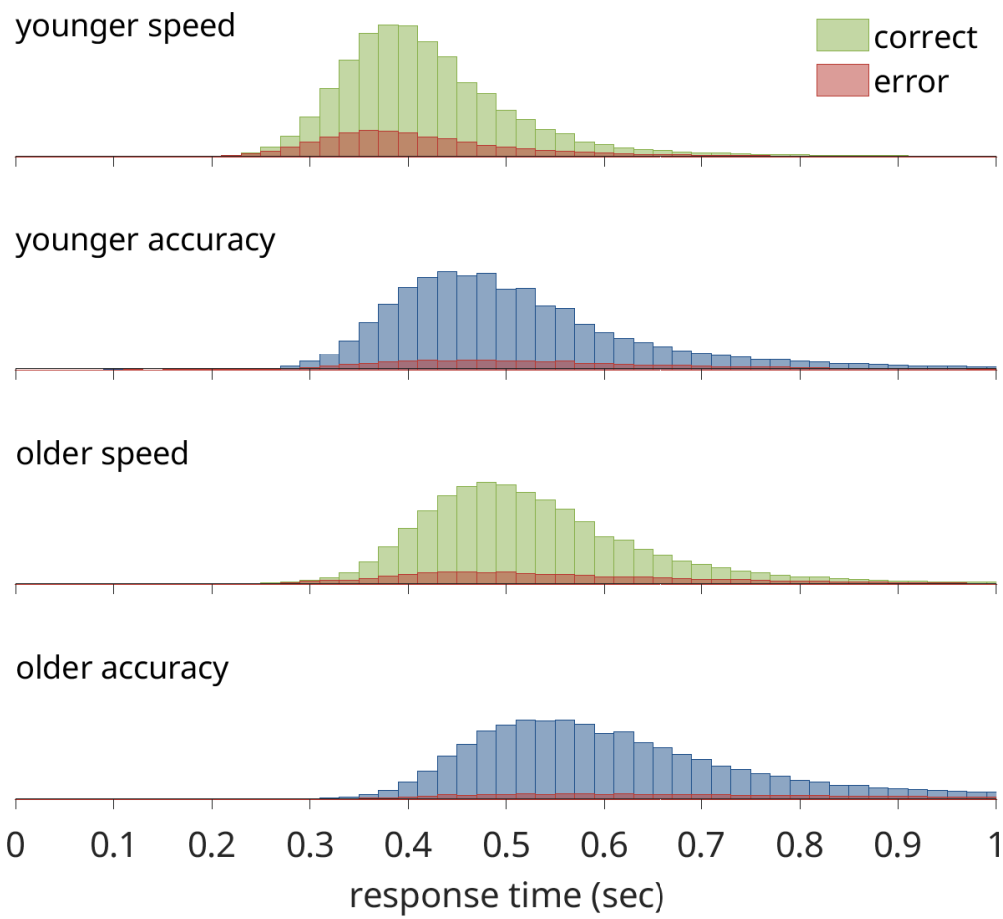
Numerosity Task

- Childers (2023) conducted a numerosity experiment
  - stimuli are visual arrays containing asterisk symbols at random location
  - had to decide whether more or fewer than 50 asterisks were present
  - used both younger and older participant groups
  - within-participant manipulation of speed versus accuracy instructions



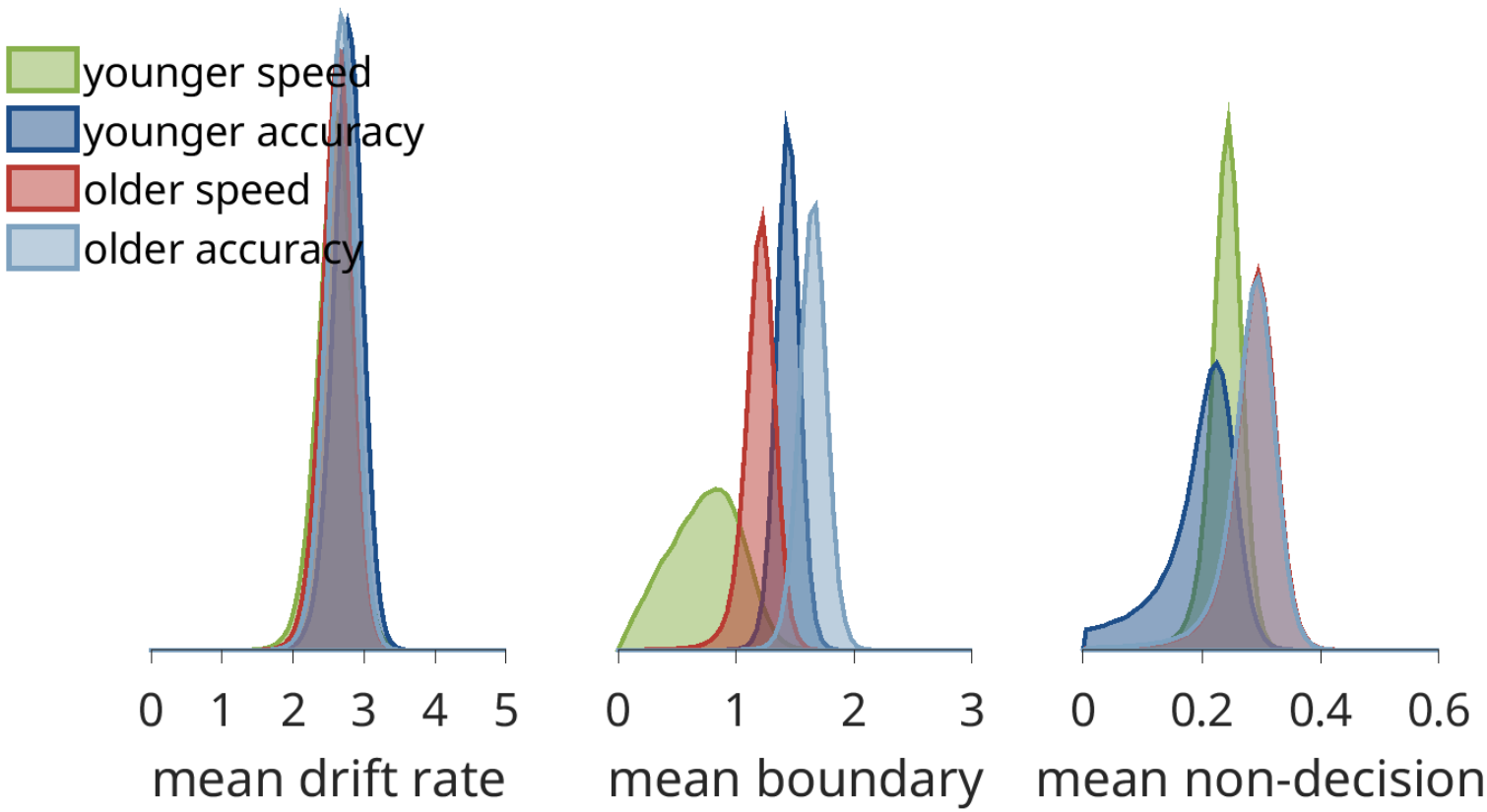
Numerosity Data

- The figure shows the decision response time distributions for four experimental groups
  - we just consider the (close to half) of trials that had easy stimuli with fewer than 40 or greater than 60 asterisks



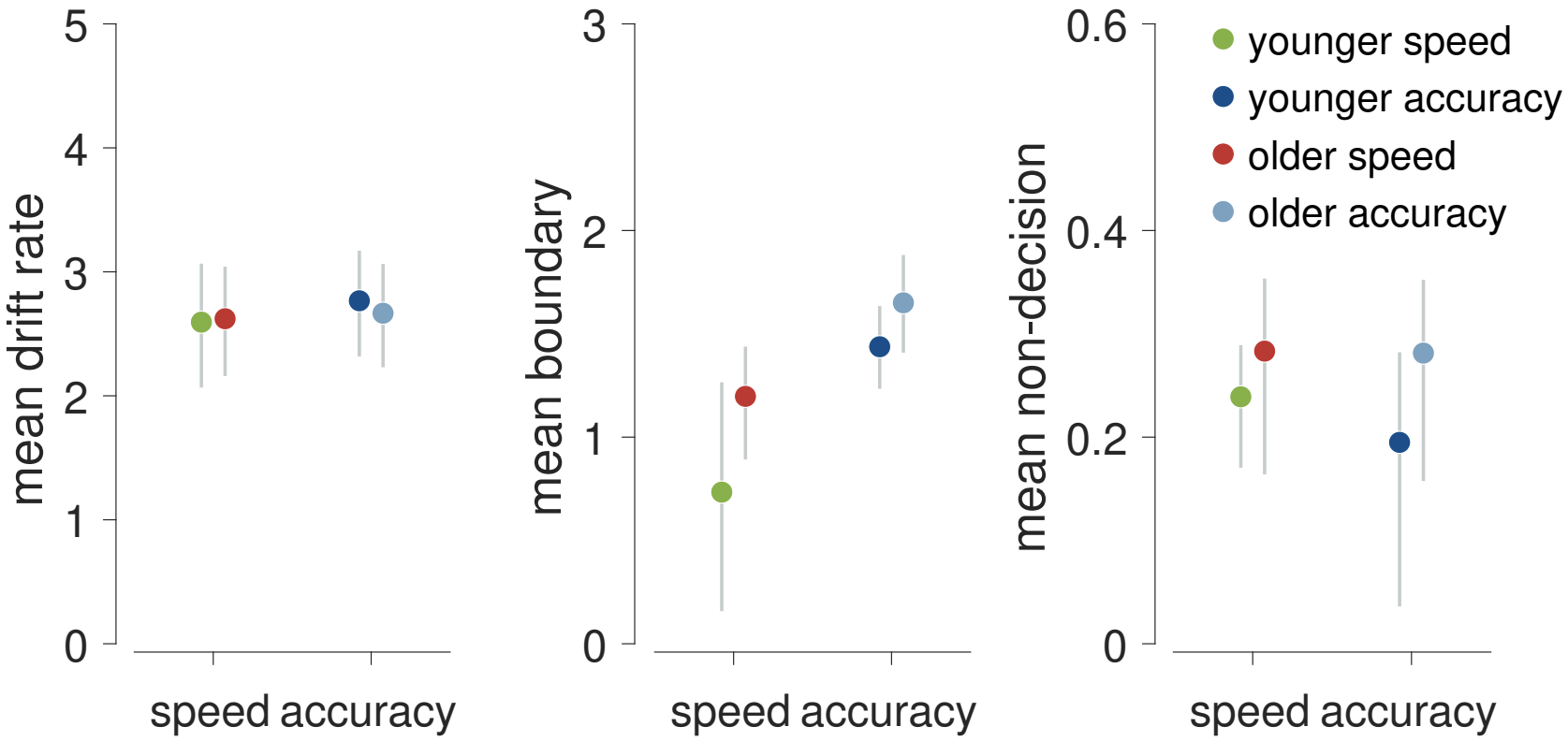
Group Inferences

- The marginal posterior distributions of the groups means for all three parameters are shown
  - drift rate does not change across groups
  - both boundary separation and non-decision time do change



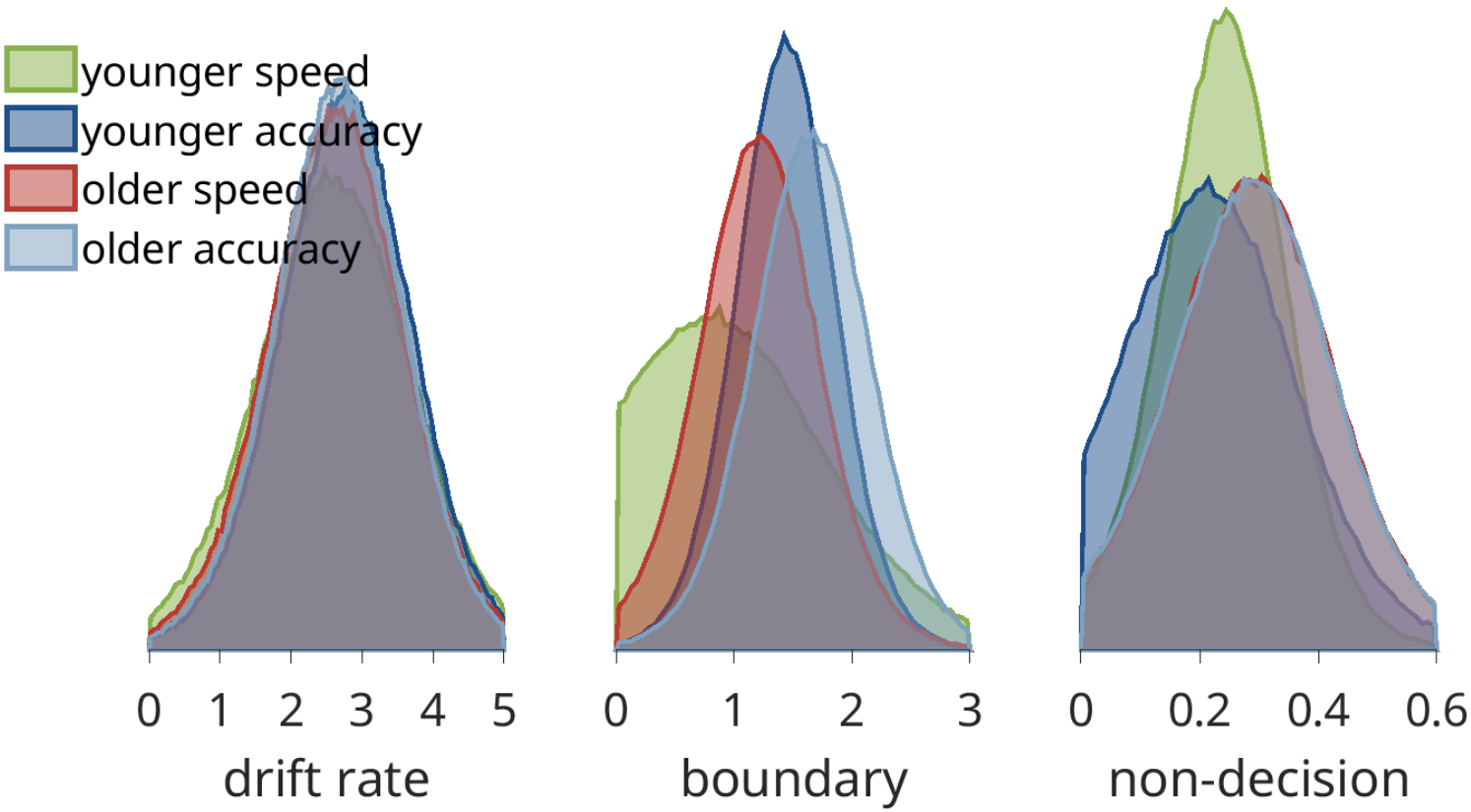
Group Inferences

- The 2×2 group structure is represented by an alternative display that summarizes the posteriors by means and 95% credible intervals
  - drift rate does not change across groups
  - the changes in boundary separation and non-decision time can be interpreted in terms of main effects and interactions



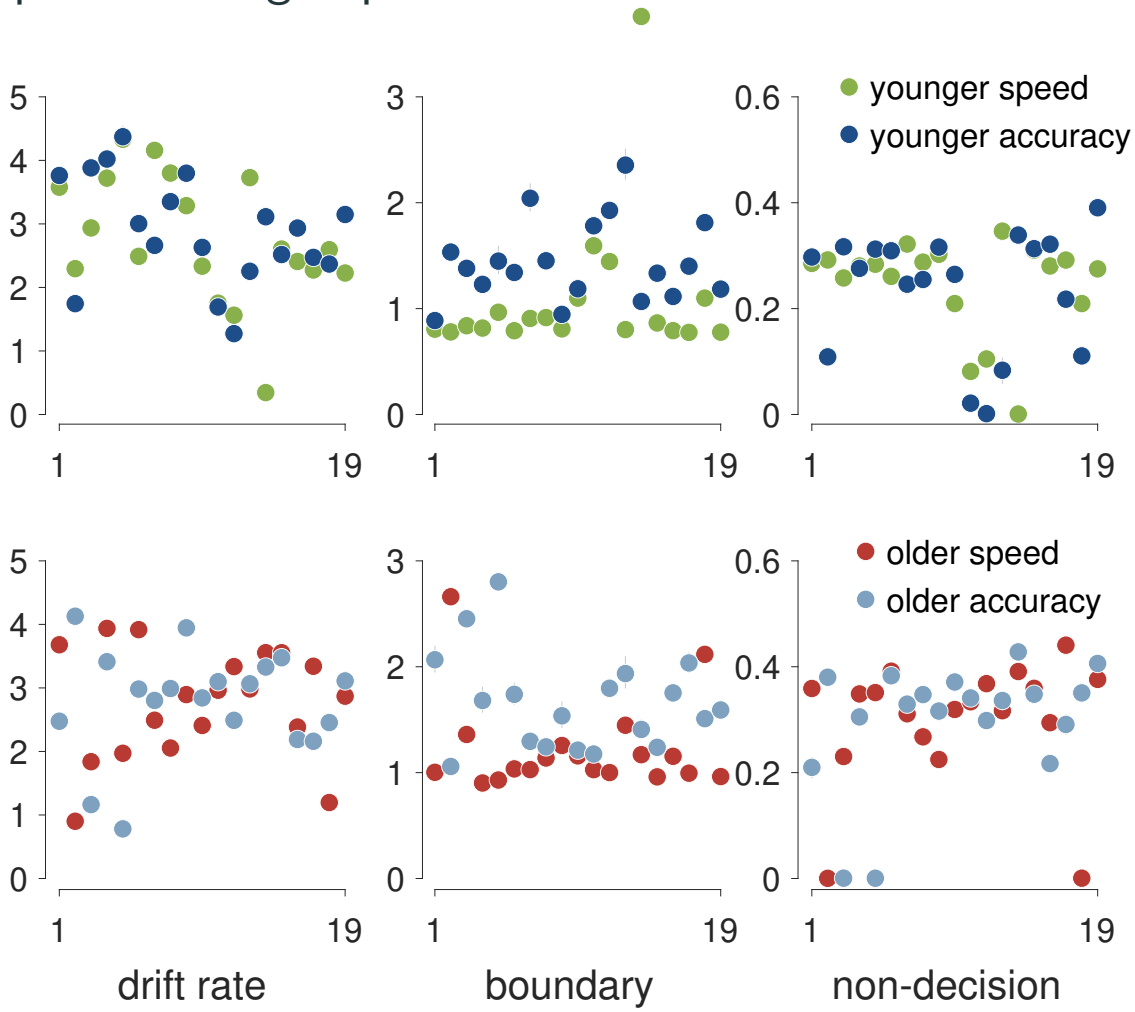
Representative Individual Inferences

- The marginal posterior distributions of the representative individual for each groups for all three parameters are shown
  - much more uncertainty, likely because of large individual differences
  - effects of age and speed versus accuracy instructions that are systematic on group means are not systematic at the individual level



Individual Inferences

- The posterior means and 95% credible intervals for each parameter of the individuals in the experiment are shown
  - shows the range of heterogeneity of the individual differences with respect to the groups



References i

Childers, R. (2023). *Speed/accuracy experiment data.* Retrieved from https://osf.io/bsvqe/